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BACKEND DEVELOPMENT ASSIGNMENT PART 6
C027-01-0863/2024
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QUESTION 1: DESIGN DECISIONS

- (i) What is your admission number, and how did it determine your dataset (number of gigs, categories, and currency)?**

My admission number is C027-01-0863/2024. Based on the assignment instructions, I created 8 gigs, used the assigned categories Marketing, Data, and Consulting, and selected KES (Kenyan Shillings) as the currency to match the project requirements.

- (ii) Why did you choose to include the status field in your gig data model? What are the possible statuses and what do they represent?**

I included the status field so that users can easily know the current progress of each freelance gig. The possible values are Open, In Progress, and Closed, which indicate whether a gig is available for applications, currently being worked on, or already completed.

Question 2: Validation

- (i) What validation rules did you apply to the budget field in your Pydantic model? Why is this validation important?**

The budget field is validated using `Field(gt=0)`, which means it must be greater than zero. This prevents invalid values such as zero or negative budgets from being accepted, ensuring that every gig has a realistic payment amount.

- (ii) How did you ensure that the category field only accepts values from your assigned category list?**

I used the Literal type in the Pydantic model to restrict the category field to Marketing, Data, and Consulting only. If a user enters any other category, FastAPI automatically rejects the request with a validation error.

Question 3: Error Handling

- (i) What status code does your API return when a user tries to retrieve a gig that does not exist? Explain why this is the appropriate status code.**

The API returns a 404 Not Found status code when the requested gig ID does not exist. This is appropriate because the resource the user requested cannot be found in the dataset.

- (ii) What status code does your API return when a user tries to create a gig with invalid data? Explain why.**

The API returns a 422 Unprocessable Entity status code when invalid data is submitted. This happens because FastAPI validates the request using the Pydantic model before creating the gig, and rejects any data that does not meet the required rules.

Question 4: Search and Filtering

- (i) Describe how your /gigs/search endpoint works. How does it handle searches where the query string does not match any gig?**

The /gigs/search endpoint searches for gigs whose titles contain the text entered by the user, regardless of letter case. If no matching titles are found, the endpoint simply returns an empty list instead of generating an error.

- (ii) Describe how your /gigs endpoint with filtering works. What are the optional query parameters, and how do they work?**

The /gigs endpoint supports optional filtering using the query parameters category, min_budget, and max_budget. These filters can be used individually or together to return only the gigs that match the selected category and budget range.

Question 5: Code Walkthrough

- (i) Write a brief explanation of the PUT /gigs/{gig_id} endpoint. What does it do, and how does it handle a request to update a gig that does not exist?**

The PUT /gigs/{gig_id} endpoint updates an existing gig by allowing changes to its budget or status. If the specified gig ID is not found, the API responds with a 404 Not Found error to inform the user that the requested gig does not exist.

- (ii) Write a brief explanation of the POST /gigs endpoint. What does it do, and how does it ensure that the new gig is created correctly?**

The POST /gigs endpoint creates a new gig by accepting validated data from the user and assigning it a unique ID before storing it in the dataset. It relies on the GigCreate Pydantic model to verify that all required fields are present and that the values satisfy the defined validation rules before the gig is added.